

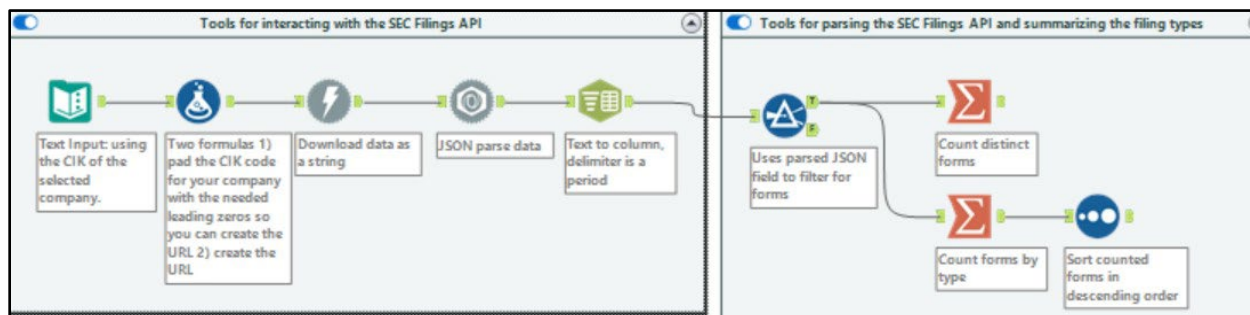
Analytics mindset

EDGAR Explorer two

Submissions Application Programming Interface (API)

Part 1 and Part 2 Alteryx workflow solution overview

Detailed instructions for each respective tool can be found in the following pages.



Part 1: Choose a company. Find its central index key (CIK) code using the Electronic Data Gathering, Analysis, and Retrieval (EDGAR) system Company Filings CIK Lookup Tool at:

<https://www.sec.gov/edgar/searchedgar/companysearch>.

As an example, for Tesla, type Tesla into the box and click "Search."

The screenshot shows the 'EDGAR | Company Filings' search page. It has a sidebar with 'Company Filings' selected. The main area has a 'Company and Person Lookup' section with a search box containing 'Tesla' and a 'Search' button. To the right is a 'How to Use this Search?' box with instructions: 'Enter name, ticker or CIK into the single search field.' and 'Suggestions as you type link directly to filings.'

The search will return the screen shown below.

Companies with names matching "TESLA"
Click on [CIK](#) to view company filings

Items 1 - 13

CIK	Company
0001728203	Tesla Auto Lease Trust 2018-A
0001345028	TESLA GROUP HOLDINGS PTY LTD
0001402995	Tesla JP/PI LLC
0001588360	Tesla Life Sciences, LLC
0001318605	Tesla, Inc. SIC: 3711 - MOTOR VEHICLES & PASSENGER CAR BODIES
0001827423	Tesla Movie, LLC
0001360796	Tesla MRI Services LLC
0001288755	Tesla Offshore L.L.C.
0001085900	TESLA POWER & AUTOMATION LP
0001085901	TESLA POWER GP INC
0001085896	TESLA POWER NEVADA INC
0001085902	TESLA POWER PROPERTIES LP
0001274730	TESLA USA INC

Depending on the company selected, there can be several different CIK codes returned, as evidenced by the Tesla example. Help your students understand that this may happen depending upon the company they select. If there are multiple companies listed, help your students understand they need to make sure

they are choosing the correct CIK. One way to confirm the company is by checking the company's identifying information in the header after clicking on the company link. For example, we can confirm that

Tesla, Inc. CIK#: 0001318605 (see all company filings) SIC: 3711 - MOTOR VEHICLES & PASSENGER CAR BODIES State location: CA State of Inc.: DE Fiscal Year End: 1231 formerly: TESLA MOTORS INC (filings through 2017-01-27) (Office of Manufacturing) Get insider transactions for this issuer.	Business Address 3500 DEER CREEK RD PALO ALTO CA 94304 650-681-5000	Mailing Address 3500 DEER CREEK RD PALO ALTO CA 94304
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Tesla's address is correct, allowing us to know we have found the correct CIK, which is 0001318605.

Part 2: Using Alteryx

Download the submissions filing for the company selected. Remember to declare your university name and university email address in the User-Agent header. [Answer: See the detailed Alteryx workbook solution for steps 1 and 2 below.](#)

- ▶ Deserialize the JavaScript Object Notation (JSON) formatted submissions filing for your company. [Answer: See the detailed Alteryx workbook solution, step 3, below.](#)

- ▶ Use the output tool to:

- Identify the number of unique forms filed by the entity. [Answer:](#) The total number of forms will vary based on the company selected. For most large companies, the answer will be around 1,000, which is the number of filings the API is expected to return. Occasionally, the API will return more than 1,000, but this should be rare.¹ You can find the detailed solution to the Alteryx workflow in steps 4 to 6 below.

1 of 1 Fields ▾ ☒ ☐	
Record	Count
1	1000

- Calculate the number of times each form was filed based on this data set and identify the most popular filing for your company. [Answer: Both the most popular filing and the number of times each form was filed will vary based on the company selected.](#) The numbers will also shift over time as the company completes different filings. Remember, only the most recent 1,000 filings are included. At the time this solution was written, for our Tesla example, the most popular filing was Form 4, which was filed 504 times. You can find the detailed solution in step 7 of the Alteryx workflow below. Form 4 is a very common filing for public companies, as are Forms 8-K. Each company's

2 of 2 Fields ▾ ☒ ☐ 39 records displayed		
Record	JSON_ValueString	Count
1	4	504
2	8-K	156
3	SC 13G/A	36
4	DEFA14A	26
5	10-Q	26
6	425	23
7	CT ORDER	23
8	424B5	20
9	PX14A6G	18
10	144	18

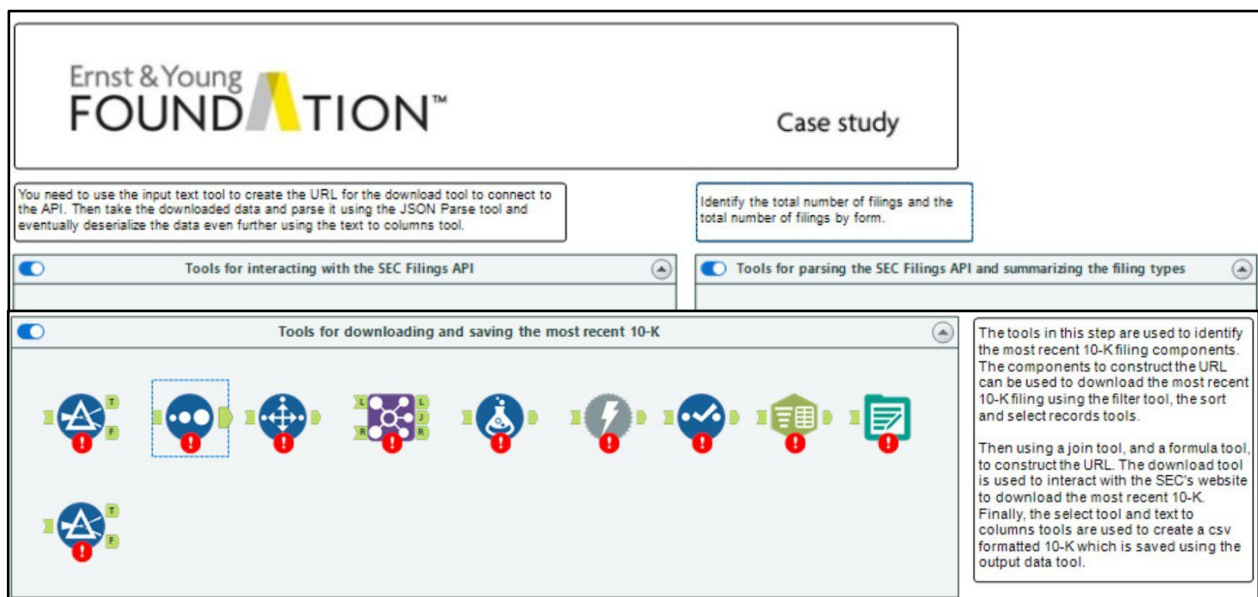
¹ According to the API documentation: "The object's property path contains at least one year's of filing or to 1,000 (whichever is more) of the most recent filings in a compact columnar data array. If the entity has additional filings, files will contain an array of additional JSON files and the date range for the filings each one contains." (See <https://www.sec.gov/edgar/sec-api-documentation>. Accessed 9/20/23)

results will differ. We recommend checking these totals for any changes the day you plan to teach this case.

Part 2: Detailed Alteryx solution

Using the starter file or a blank Alteryx workflow

This case includes a starter file that you may wish to provide to your students. The file is labeled **Analytics_mindset_case_study_EDGAR_Explorer_two_Submissions_API_starter.yxmd** and includes prepopulated tools demonstrating how to create the workflow, as well as commentary about the various tools and steps. The case can also be taught using a blank Alteryx workflow. Note that the starter file will display errors upon opening as there is no input data or connections between tools and also because each tool is not configured. The students will need to work with the file before the workflow will run anything. For example, the provided starter file looks like the screenshot below.



Solution

The solution is written using a blank Alteryx workflow. If you choose to use the starter file, skip the steps outlining which tool to drag into the workflow and start at the configuration portion of the instructions.

Download the submissions filing for the company you selected. Remember to declare your university name and university email address in the User-Agent header.

► Step 1: Text Input tool (extract, transform, load (ETL) steps)

Enter the CIK of the selected company.

- Drag a **Text Input** tool onto the workflow.
- In the configuration window, change the header to cik and type in the company's central index key (CIK) in row one.

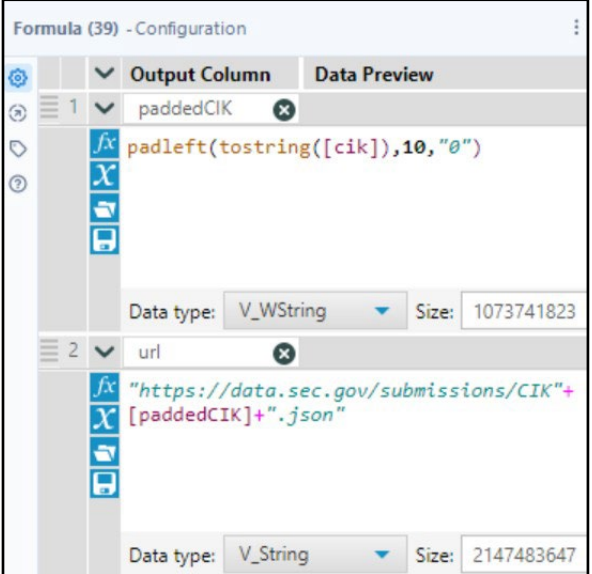


	cik
1	1318605

► Step 2: Formula tool

You need to change the format of the CIK entered in the previous step to a string format in Alteryx and pad the CIK number with the right number of leading zeros per the SEC format.

- Drag a **Formula tool** onto the workflow.
- Create a new column to hold the paddedCIK. In the configuration window, change the output column to paddedCIK and enter the formula as shown. This will take the CIK code provided in step 1 and add leading zeros (0) up to 10 characters (length), as specified by the formula. The padleft formula is string, length, character.
- Using the same formula tool, open up the window below your first formula by clicking the blue plus symbol. Configuration space 2 will open, allowing you to input the URL using the newly created padded CIK.



Formula (39) - Configuration		
	Output Column	Data Preview
1	paddedCIK	<code>padleft(tostring([cik]),10,"0")</code> Data type: V_WString Size: 1073741823
2	url	<code>"https://data.sec.gov/submissions/CIK"+[paddedCIK]+".json"</code> Data type: V_String Size: 2147483647

- Enter url as the name of your new output column.
- Type in the EDGAR URL: "https://data.sec.gov/submissions/CIK"+[paddedCIK]+".json."

► **Step 3: Download tool (ETL)**

- Drag a **Download** tool onto the workflow and connect it to the **Formula** tool from the prior step.
- In the configuration window, select the **Basic** header (should be the default).
 - The Data Connection Manager should be unchecked.
 - In the **URL** field section, select the variable “url” from the drop-down menu and ensure the encode url text is checked.
 - In the Output section, select “string” in the To a Field section.
- Select the **Headers** header in the configuration window.
 - Click the “Add” button.
 - Fill in **Name** with “user-agent.”
 - Fill in **Value** with the name of your university and your university email address, as shown below.
 - Do not alter **Payload** and **Connection** headers for this case.



The screenshot shows the configuration window for the Download tool, with the 'Basic' header selected. The 'URL' field is set to 'url' and 'Encode URL Text' is checked. Under the 'Output' section, 'To a Field' is selected with 'String' as the output type and 'Unicode UTF-8' as the encoding. The 'To a File' section is collapsed.

The screenshot shows the 'Headers' tab of the configuration window. It displays a table for adding constant values:

Name	Value
user-agent	Your University you@your.edu

Buttons for 'Add' and 'Delete' are visible on the right.

This screenshot shows the 'Headers' tab with the 'user-agent' header successfully added to the table.

Deserialize the JSON formatted submissions filing for your company.

► **Step 4: JSON Parse tool (ETL)**

- Drag a **JSON Parse** tool onto the workflow and connect it to the **Download tool**.
- In the configuration window, select the “DownloadData” variable in the JSON Field drop-down menu,
- Select **Output values into single string field**.



The screenshot shows the configuration window for the JSON Parse tool. The 'JSON Field' is set to 'DownloadData'. The 'Include in Output' checkbox is unchecked. The output option 'Output values into single string field' is selected.

- You may find it useful to have the students examine the deserialized JSON, noting the variable JSON_Name includes an index linking all the information about the filings. The JSON_Name field will include both header information and an index of the filings. For example, in the Tesla download:
 - ▶ The header information ends with information about Tesla's former names and the time during which these names were active. For example, for the JSON_Name formerNames.0.name, the corresponding JSON_ValueString is TESLA MOTORS INC, which was Tesla's former name.
 - ▶ The filings information begins after the header information with the JSON_Name filings.recent.accessionNumber.0. In this example, the JSON_ValueString is 0001628280-23-034847, which is the accession number of the most recent filing as of October 23, 2023. This number will change each time Tesla makes a filing. The exact location of the first accession number will vary between companies depending on how much information is included in the header, but in many cases this information will be found after row 30, which will often be the company's phone number, depending on if there are any flags or name changes. In our Tesla example, the first accession number can be found in row 35.

Record	DownloadHeaders	JSON_Name	JSON_ValueString
25	HTTP/1.1 200 OK	addresses.business.street2	[Null]
26	HTTP/1.1 200 OK	addresses.business.city	PALO ALTO
27	HTTP/1.1 200 OK	addresses.business.stateOrCountry	CA
28	HTTP/1.1 200 OK	addresses.business.zipCode	94304
29	HTTP/1.1 200 OK	addresses.business.stateOrCountryDescription	CA
30	HTTP/1.1 200 OK	phone	650-681-5000
31	HTTP/1.1 200 OK	flags	
32	HTTP/1.1 200 OK	formerNames.0.name	TESLA MOTORS INC
33	HTTP/1.1 200 OK	formerNames.0.from	2005-02-17T00:00:00.000Z
34	HTTP/1.1 200 OK	formerNames.0.to	2017-01-27T00:00:00.000Z
35	HTTP/1.1 200 OK	filings.recent.accessionNumber.0	0001628280-23-034847
36	HTTP/1.1 200 OK	filings.recent.accessionNumber.1	0001628280-23-034588
37	HTTP/1.1 200 OK	filings.recent.accessionNumber.2	0001771340-23-000009
38	HTTP/1.1 200 OK	filings.recent.accessionNumber.3	0000950170-23-052618
39	HTTP/1.1 200 OK	filings.recent.accessionNumber.4	0001950047-23-004427

Use the output tool to:

- ▶ Identify the number of unique forms filed by the entity. (Steps 5 to 7)
- ▶ Calculate the number of times each form was filed based in this data set and identify the most popular filing for your company. (Step 8)

▶ **Step 5: Text to Columns tool (ETL)**

- Drag a **Text to Columns** tool onto the workflow and connect it to the **JSON Parse** tool.
- In the configuration window, select the “JSON_Name” variable in the *Column to split* drop-down menu.
- Add a period (.) as the delimiter.
- Select “Split to columns” and increase the *Number of columns* to 4.
- For Extra characters, select “Leave extra in last column.”
- For Output root name, type in JSON_Name (which is expected to be the default).



▶ **Step 6: Filter tool (ETL)**

Using the JSON_Name as the output root name in step 4 above creates four output variable JSON_Name fields appended with the numbers 1 through 4. The third variable can be used to find all of the forms filings. Reviewing the file and visually seeing the columns will help you know which fields to select and sort. For this step, review the output from step 4 and look at the various fields. You'll notice that JSON_Name3 has form data and the corresponding JSON_ValueString will have the type of form filed.

- Drag a **Filter** tool onto the workflow and connect it to the **Text to Columns** tool.
- In **Basic filter**, select the column name “JSON_Name3.”
 - ▶ Choose “Equals.”
 - ▶ Type form.



► **Step 7: Summarize tool to identify the number of unique forms filed by the entity**

- Drag a **Summarize** tool onto the workflow.
- Connect it to the **T** (true) output node of the **Filter** tool.
- In the **Fields** section, select the variable “JSON_Name,” then in the **Action** section, use the Add drop-down menu to select “**Count Distinct.**”



Field	Type
url	String
DownloadHeaders	V_String
JSON_Name	V_WString
JSON_ValueString	V_WString
JSON_Name1	V_WString
JSON_Name2	V_WString
JSON_Name3	V_WString
JSON_Name4	V_WString

Field	Action	Output Field Name
JSON_Name	Count Distinct	CountDistinct_JSON...

Part 2 answer: There are 1,000 unique forms, but this number will vary and can be above 1,000 for large public companies and could be below 1,000 for recently listed companies (see the explanation in an earlier answer, page 2 of this solution guide and in footnote 1).

1 of 1 Fields ☒ ☐

Record	Count
1	1000

► **Step 8: Summarize and Sort tools**

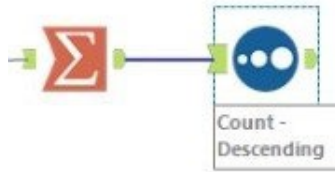
- We need to calculate the total number of times each form was filed and identify the most popular filing for your company. You’ve already calculated the total number of forms, now you just need to summarize those forms by type, which is contained in the JSON_ValueString field.
- Drag a second **Summarize** tool onto the workflow. Connect it to the same **T** (true) output node of the **Filter** tool used in step 5 (the filter tool should now have two outgoing summarize connections).
- In the **Fields** section of the configuration window, select “JSON_ValueString” and use the Add drop-down menu in the **Actions** section to select “**Group By.**”
- In the **Fields** section, select “JSON_ValueString” again, but this time use the Add drop-down menu to select “**Count**” in **Action**.

Field	Type
url	String
DownloadHeaders	V_String
JSON_Name	V_WString
JSON_ValueString	V_WString
JSON_Name1	V_WString

Field	Action	Output Field Name
JSON_ValueString	Group By	JSON_ValueString
JSON_ValueString	Count	Count

- Drag a **Sort** tool onto the workflow and connect it to the second **Summarize** tool (the one just configured).

- ▶ Under **Name**, select the variable “Count” and under **Order**, select “Descending.”



Solution to: Calculate the number of times each form was filed based in this dataset and identify the most popular filing for your company.

- ▶ The most popular filing will generally be an insider trading form, i.e., Form-4. Remember the totals will change from the answer posted here as the company submits additional filings to the SEC.

☒ Use Dictionary Order English (United States)

Fields

	Name	Order
▶	Count	Descending
*		

2 of 2 Fields ☒ 39 records displayed

Record	JSON_ValueString	Count
1	4	504
2	8-K	156
3	SC 13G/A	36
4	DEFA14A	26
5	10-Q	26
6	425	23
7	CT ORDER	23
8	424B5	20
9	PX14A6G	18
10	144	18

Part 3: Researching different types of forms or SEC filings

- ▶ Based on your analysis above, pick two forms you have not heard of before and research the purpose of those forms. The SEC website <https://www.sec.gov/forms> provides a useful starting point for your research. A filing with “/A” appended to the end is an amended filing. Amended filings are corrected filings of forms that were previously filed with the SEC.

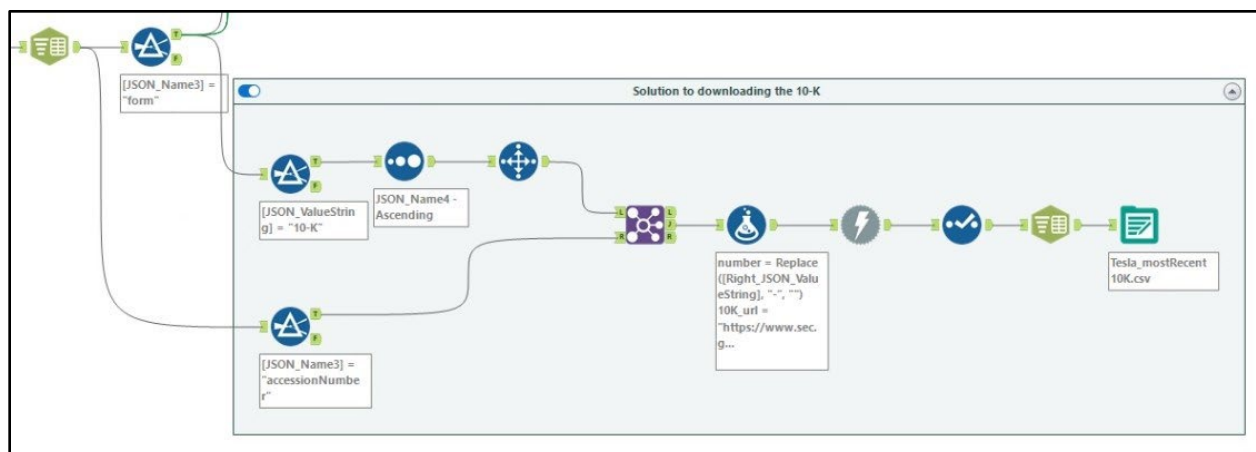
Write a brief description of the forms you are researching, including:

- (i) What information is being disclosed by the company.
 - (ii) Why this information is important to users of financial information.
- ▶ Return to your Alteryx workflow. Use the appropriate tools to identify the most recent Form 10-K and use the `accessionNumber` with the SEC EDGAR Archive data URL <https://www.sec.gov/Archives/edgar/data> and the company’s CIK to download the most recent Form 10-K and save it as a csv file. The website address that resolves the filing will be a text document, with the extension `.txt`.
 - ▶ Save your Alteryx workflow and submit it for grading.

Part 3 Alteryx solution

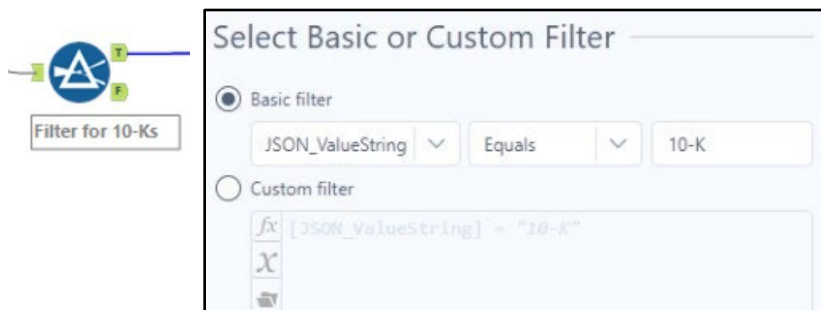
This portion of the Alteryx workflow has two incoming nodes, both of which link to different **Filter** tools in the workflow.

- ▶ The first **Filter** tool is used to identify the company’s Form 10-K filings and connects to the **Filter** tool configured in Part 2, step 5 of this case.
- ▶ The second **Filter** tool is used to identify accession numbers and connects to the **Text to columns** tool configured in Part 2, step 4 of this case. See below.

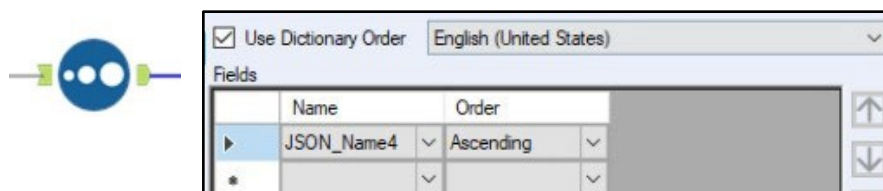


► **Step 1: Filter, Sort and Select Records tools to identify the most recent Form 10-K filing**

- Drag a **Filter** tool onto the workflow and connect it to the T (true) value of the previous **Filter** tool from Part 2, step 5.
- In **Basic filter**, Select “JSON_ValueString,” then “Equals” and then type in 10-K.



- Drag a **Sort** tool onto the workflow and connect it to the T (true) value of the new Filter tool just configured.
- Under **Name**, select the variable “JSON_Name4.”
- Under **Order**, select “Ascending” (JSON_Name4 is the index number for the filings). The first record is now the latest Form 10-K filed.



- Drag a **Select Records** tool onto the workflow.
- Connect it to the Sort tool just configured.
- Under **Ranges**, type in 1, which will display the first row only, or in this case, the last filing.

Questions

Enter the numeric ranges of records to return. For Example:

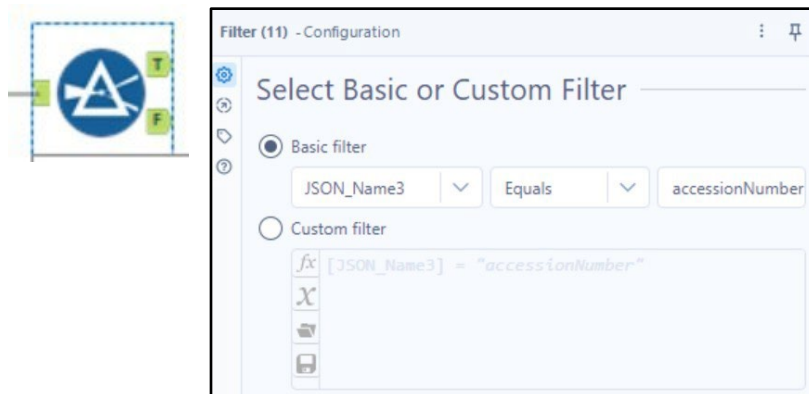
-2
3
17-20
50+

Ranges:

1

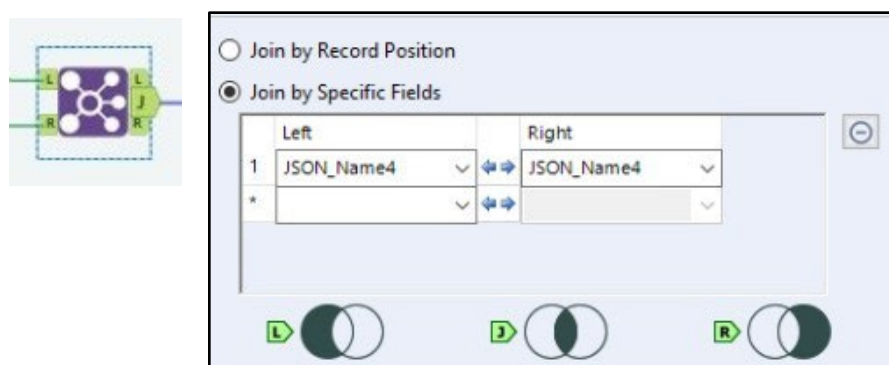
► **Step 2: Filter tool to keep only accession numbers from all filing data**

- Drag a **Filter** tool onto the workflow and connect it to the **Text to Columns** tool created in Part 2, step 4 of this case.
- In **Basic filter**, select “JSON_Name3,” then equals and then type in accessionNumber.



► **Step 3: Join tool to match the most recent Form 10-K with the correct accession number**

- Drag a **Join** tool onto the workflow.
- Connect the outgoing node from the **Select Records** tool in step 1 of Part 3 to the incoming **L** node and the outgoing T (true) node from the **Filter** tool in step 2 of Part 3 to the incoming Join node **R**.
- Select **Join by Specific Fields**.
 - Under **Left**, select the variable “JSON_Name4.”
 - Under **Right**, select the variable “JSON_Name4.”



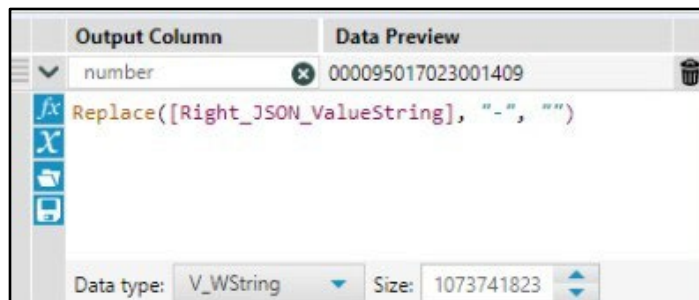
► **Step 4: Formula tool to create the EDGAR archives URL for the Form 10-K filing**

- Drag a **Formula** tool onto the workflow.
- Connect the **Formula** tool to the outgoing **J** node of the **Join** tool added in Part 3, step 3.
- To construct the EDGAR URL for the Form 10-K, we need to modify the incoming data and output a correct URL. The correct URL includes a standard part, the accession number

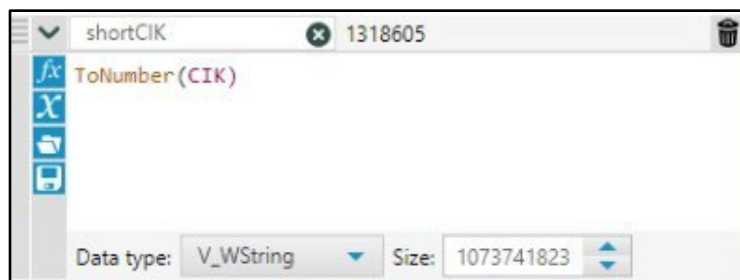


without dashes and the CIK without the leading zeros. A sample URL is provided for you as you work through the following steps.

- ▶ There are three formulas needed to create the URL. You can use one **Formula** tool to add all four formulas to the workflow. For all three of these formulas, you will create a new output column with the results of the formula. The steps to do this are:
 - Under the Output Column in the configuration window, choose “select column.”
 - From the drop-down menu, choose “+Add Column” from the *Select Column* drop-down menu. Type in the new column name.
 - First formula: Remove the dashes from the accession number.
 - ▶ Create a new column as instructed. Title the new column “number” (without quotes).
 - ▶ Type in the formula: `Replace([Right_JSON_ValueString], “-”, “”)`.

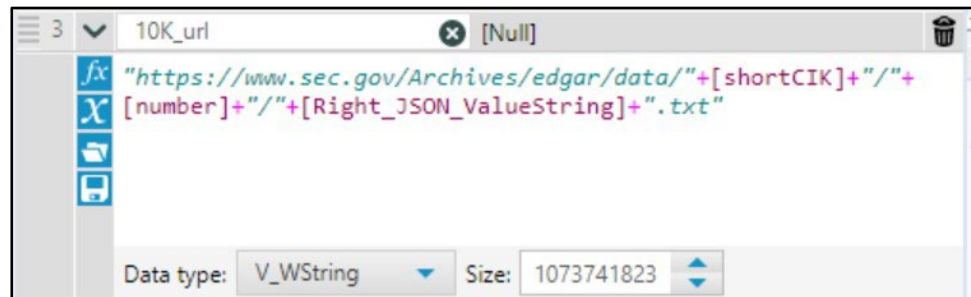


- Second formula: Remove the leading zeros. Do not use a replace in case there are zeros in the CIK.
 - ▶ Stay in the same Formula tool. Under the box containing the previous formula, click on the blue + symbol to open another configuration box.
 - ▶ Create a new column as instructed. Title the new column “shortCIK” (without quotes).
 - ▶ Enter the formula: `ToNumber([CIK])`.
 - ▶ Do not change the *Data type* to a numerical format. Keep the result a string.

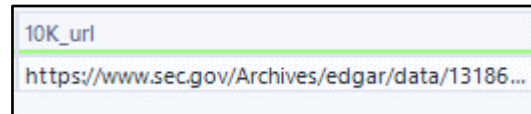


- Third formula: Combine the newly created information to create the unique URL that will enable you to download the most recent Form 10-K filing for the company selected.
- ▶ Stay in the same Formula tool. Under the box containing the previous formula, click on the blue + symbol to open another configuration box.
- ▶ Create the new column as instructed, naming the column “10k_url” (without quotes).
- ▶ Enter the formula below. Note that Archives must be capitalized for the download to function.

“https://www.sec.gov/Archives/edgar/data/”+[shortCIK]+”/”+[number]+”/”+[Right_JSON_ValueString]+”.txt”

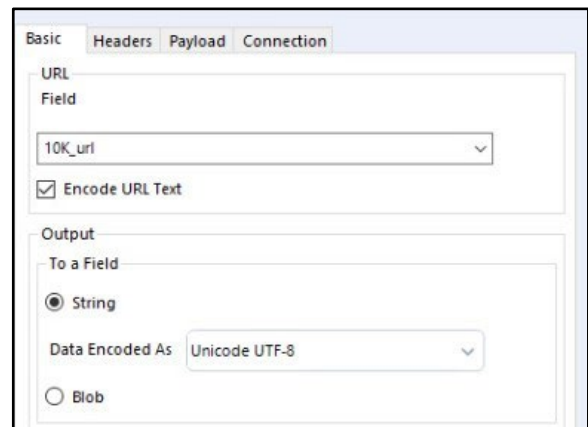
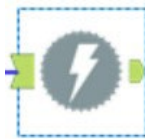


- ▶ In this case, the correct url will be:
<https://www.sec.gov/Archives/edgar/data/1318605/000095017023001409/0000950170-23-001409.txt>.
- ▶ The bold and italic portions of the URL above will change as the firm files a new form.²



▶ Step 5: Download tool to download the most recent Form 10-K filing

- Drag a **Download** tool onto the workflow.
- Connect the **Download** tool to the **Formula** tool from the prior step.
- Select the **Basic** header in the configuration window (should be the default).
- ▶ In the URL section, select the variable url (10K_url) in the *Field* drop-down menu.
- ▶ In the *Output* section, select **String** from the *To a Field* section.



² Alteryx’s preview may show as [Null] as the resulting url string is very long. Alteryx will, however, still compute the correct URL, as shown in our Tesla example.

- Select the *Headers* tab in the configuration window.

- ▶ Click the “Add” button and fill in *Name* with User-Agent and *Value* with the name of your university and your university email address. Both the User-Agent and the details should be without quotation marks (same as Part 2, step 2).
- ▶ The *Payload* and *Connection* headers do not need to be altered for this case.

The screenshot shows the 'Headers' tab in a configuration window. It has a sub-header 'Add from the following constant values'. Below this is a table with two columns: 'Name' and 'Value'. To the right of the table are two buttons: 'Add' and 'Delete'.

The screenshot shows the 'Headers' tab with the 'user-agent' header added. The table now has one row with 'user-agent' in the 'Name' column and 'Your University you@your.edu' in the 'Value' column. The 'Add' and 'Delete' buttons are still present.

▶ **Step 6: Select tool to choose only the field you want to continue using in the workflow**

- Connect the **Select** tool to the **Download** tool from the step 5.
- Deselect all fields, except for the variable DownloadData. This field contains the Form 10-K filing in a raw text format.
- In the picture below, the DownloadData field has been moved from the bottom of field list in the configuration window to the top. You can do that by using the arrows at the top of the field list, next to *Options*.



The screenshot shows the 'Select (18) - Configuration' window. It has a table with columns: 'Field', 'Type', 'Size', and 'Rename'. The 'DownloadData' field is selected (checked) and moved to the top of the list. Below it are other fields like 'URL', 'DownloadHeaders', 'JSON_Name', etc. There are also 'Options' and 'Search' buttons at the top.

Field	Type	Size	Rename
☒ DownloadData	V_WString: Forced	107...	
☐ URL	V_String	214...	
☐ DownloadHeaders	V_String	214...	
☐ JSON_Name	V_WString	107...	
☐ JSON_ValueString	V_WString	107...	
☐ JSON_Name1	V_WString	107...	
☐ JSON_Name2	V_WString	107...	
☐ JSON_Name3	V_WString	107...	
☐ JSON_Name4	V_WString	107...	
☐ Right_URL	V_String	214...	
☐ Right_DownloadHeaders	V_String	214...	

► **Step 7: Text to columns tool to split the URL text file you created into a CSV file**

- Connect the **Text to columns** tool to the **Select** tool from the prior step.
- Under *Column to split*, select “DownloadData” (it should be the only option following the use of the **Select** tool in the prior step).
- Under Delimiters, type in “\n,” which designates you want to split the text file into a CSV at the end of each line of text.
- Select “Split to rows.”



Select Column to Split

Column to split: Delimiters:

☐ Split to columns

☒ Split to rows

Advanced options

☐ Ignore delimiters in quotes

☐ Ignore delimiters in single quotes

☐ Ignore delimiters in parentheses

☐ Ignore delimiters in brackets

☐ Skip empty columns

► **Step 8: Output data**

- Connect an **Output data** tool to the **Text to columns** tool from the prior step.
- Under *Write to File or Database*, type in the name of the file you wish to save, followed by the extension .csv. For our Tesla example, **Analytics_mindset_case_studies_solutions_EDGAR_Explorer_two_Submissions_API_Tesla_mostRecent10K.csv**, make sure Use Data Connection Manager is unchecked.
- To make the output data work, your Alteryx file needs to be saved. When you run the workflow, the output file will be in the same folder where you saved your Alteryx workbook.
- Be aware that the returned data will be in HTML language and these documents include a large amount of formatting information.

Write to File or Database

Options

Name	Value
1 Max Records Per File	
2 File Format	Comma Separated Value (*.csv)
3 Delimiters	,
4 First Row Contains Field Names	☒
5 Quote Output Fields	Auto
6 Code Page	ISO 8859-1 Latin I
7 Line Ending Style	Windows
8 Write BOM	☒